Ariq Fadhilah Jeha

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Background

I am a Biomedical Engineering student with a strong interest in Integrated Circuit (IC) design, spanning both analog and digital domains. I have hands-on experience in both analog and digital , from functional verification untill post-layout simulation using Cadence.

Education

Gadjah Mada University, Biomedical Engineering — August 2022 – Present

- GPA: 3.39/4.0
- **Coursework:** Fundamental Electronics, Biomedical Electronics, Digital Engineering, Microcontroller System , VLSI Development, Digital System Development.

Experience

Research Intern, Beehive Drones – Jogjakarta — Jan 2025 – Feb 2025

- Developed a FAST keypoint detection module using Vitis HLS.
- Designed and verified the accelerator at C/C++ and HLS simulation level.

Research Assistant, DTETI – Jogjakarta — May 2025 – May 2026

- Developing custom AXI4-Lite peripherals for a RISC-V-based microcontroller, including SPI and UART interfaces.
- Designed and verified Verilog RTL using simulation and custom testbenches.

Projects

1. Phase-Locked Loop In MICS Band

- Designed and implemented a full-custom Integer-N PLL in CMOS GPDK045 for 402–405 MHz MICS-band applications, including PFD, charge pump, loop filter, current-starved VCO, and frequency divider.
- Completed full-custom layout, DRC/LVS verification, parasitic extraction, and post-layout simulation, achieving 402–405 MHz output with 15–39 μs settling time and 214.85–230.2 μW power consumption.

2. Ambulatory ECG System

- Developed an ambulatory ECG acquisition system using STM32F4 (300 Hz sampling rate).
- Implemented timer-driven UART acquisition with interrupt + DMA and circular buffering.
- Transmitted real-time ECG data via Bluetooth.

3. AXI4-Lite SPI & UART Peripheral Design

- Designed SPI master and UART communication peripherals.
- Implemented AXI4-Lite wrappers with memory-mapped control and status registers.
- Integrated FIFO buffering for UART data handling.
- Verified functionality through RTL simulation and testbench-based validation.

Technologies

Languages: Tcl, Verilog, SystemVerilog, Python, C++

EDA & Tools: Cadence Virtuoso, Spectre ADE, Genus, Innovus, Tempus, Xilinx Vivado, Vitis

Environment: Linux (CLI, shell scripting)